

```
docker run -d -p 8080:8080 --name airflow apache/airflow standalone
```

```
docker ps
```

```
docker logs airflow
```

```
docker exec -it airflow bash
```

```
cd /opt/airflow/dags
```

```
cat > daily_attendance_processor.py
```

```
from airflow import DAG
```

```
from airflow.operators.python import PythonOperator
```

```
from datetime import datetime
```

```
ATTENDANCE_FILE = "/tmp/attendance.txt"
```

```
REPORT_FILE = "/tmp/attendance_report.txt"
```

```
def create_attendance_file():
```

```
    students = [
```

```
        "Aarav,Present",
```

```
        "Priya,Present",
```

```
        "Rahul,Absent",
```

```
        "Sneha,Present",
```

```
        "Kiran,Absent",
```

```
        "Ananya,Present",
```

```
        "Vikram,Present",
```

```
        "Meera,Absent",
```

```
        "Farhan,Present",
```

```
        "Divya,Present"
```

```
    ]
```

```
with open(ATTENDANCE_FILE, "w") as file:
    for student in students:
        file.write(student + "\n")
    print("Attendance file created successfully")

def read_attendance_file():
    with open(ATTENDANCE_FILE, "r") as file:
        data = file.readlines()
    print("Attendance Records:")
    for line in data:
        print(line.strip())

def count_total_students():
    with open(ATTENDANCE_FILE, "r") as file:
        data = file.readlines()
    total_students = len(data)
    print(f"Total Students = {total_students}")

def count_present_students():
    present_count = 0
    with open(ATTENDANCE_FILE, "r") as file:
        for line in file:
            name, status = line.strip().split(",")
            if status == "Present":
                present_count += 1
    print(f"Present Students = {present_count}")

def count_absent_students():
    absent_count = 0
    with open(ATTENDANCE_FILE, "r") as file:
        for line in file:
            name, status = line.strip().split(",")
```

```

        if status == "Absent":

            absent_count += 1

    print(f"Absent Students = {absent_count}")

def calculate_attendance_percentage():

    total_students = 0

    present_students = 0

    with open(ATTENDANCE_FILE, "r") as file:

        for line in file:

            total_students += 1

            name, status = line.strip().split(",")

            if status == "Present":

                present_students += 1

    percentage = (present_students / total_students) * 100

    print(f"Attendance Percentage = {percentage:.0f}%")

def list_absent_students():

    print("Absent Students List")

    with open(ATTENDANCE_FILE, "r") as file:

        for line in file:

            name, status = line.strip().split(",")

            if status == "Absent":

                print(name)

def generate_attendance_report():

    total_students = 0

    present_students = 0

    absent_students = 0

    with open(ATTENDANCE_FILE, "r") as file:

        for line in file:

            total_students += 1

```

```

        name, status = line.strip().split(";")

        if status == "Present":

            present_students += 1

        else:

            absent_students += 1

    percentage = (present_students / total_students) * 100

    if percentage >= 75:

        status = "Good"

    else:

        status = "Needs Improvement"

    report = f"""
Daily Attendance Report
Total Students = {total_students}
Present Students = {present_students}
Absent Students = {absent_students}
Attendance Percentage = {percentage:.0f}%
Status = {status}
"""

    with open(REPORT_FILE, "w") as file:

        file.write(report)

    print("Attendance report generated successfully")

with DAG(
    dag_id="daily_attendance_processor",
    start_date=datetime(2024, 1, 1),
    schedule=None,
    catchup=False
) as dag:

```

```
create_attendance_file_task = PythonOperator(
    task_id="create_attendance_file",
    python_callable=create_attendance_file
)

read_attendance_file_task = PythonOperator(
    task_id="read_attendance_file",
    python_callable=read_attendance_file
)

count_total_students_task = PythonOperator(
    task_id="count_total_students",
    python_callable=count_total_students
)

count_present_students_task = PythonOperator(
    task_id="count_present_students",
    python_callable=count_present_students
)

count_absent_students_task = PythonOperator(
    task_id="count_absent_students",
    python_callable=count_absent_students
)

calculate_attendance_percentage_task = PythonOperator(
    task_id="calculate_attendance_percentage",
    python_callable=calculate_attendance_percentage
)

list_absent_students_task = PythonOperator(
    task_id="list_absent_students",
    python_callable=list_absent_students
)
```

```
generate_attendance_report_task = PythonOperator(  
    task_id="generate_attendance_report",  
    python_callable=generate_attendance_report  
)
```

```
create_attendance_file_task >> read_attendance_file_task >>  
count_total_students_task >> count_present_students_task >>  
count_absent_students_task >> calculate_attendance_percentage_task >>  
list_absent_students_task >> generate_attendance_report_task
```

```
airflow@e25d12bf1571:/opt/airflow/dags$ cat /tmp/attendance.txt
```

Aarav,Present

Priya,Present

Rahul,Absent

Sneha,Present

Kiran,Absent

Ananya,Present

Vikram,Present

Meera,Absent

Farhan,Present

Divya,Present

```
airflow@e25d12bf1571:/opt/airflow/dags$ cat /tmp/attendance_report.txt
```

Daily Attendance Report

Total Students = 10

Present Students = 7

Absent Students = 3

Attendance Percentage = 70%

Status = Needs Improvement

exit

docker restart airflow

The screenshot shows the Apache Airflow web interface in a browser. The address bar displays the URL: `b67ac309ed02-10-244-8-210-8080.spc.a.r.killercoda.com/dags`. The interface has a dark theme and a sidebar on the left with navigation links: Home, Dags (selected), Assets, Browse, Admin, Docs, and a clock icon showing 10:53. The main content area is titled 'Dags' and includes a search bar and a 'Ctrl+K' shortcut. Below the search bar, there are tabs for 'All', 'Failed', 'Queued', 'Running', 'Success', and 'Required Actions'. The 'All' tab is selected, and it shows a list of dag status filters: 'All', 'Active', 'Paused', and 'Filter by tag'. A dropdown menu is open, showing 'All', 'Favorite', and 'Unfavorite'. The list of dags shows '1 Dag' and a table with the following data:

Schedule	Latest Run	Next Run
	2026-05-09 12:22:11 ✓	

The 'daily_attendance_processor' dag is highlighted in blue. To the right of the dag name, there are icons for a toggle switch, a play button, a star, and a trash can. The 'Latest Run' column shows a green checkmark, indicating a successful run.